

COMP4616 – Fundamentals of Machine Learning

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SPRING 2026

WEEK 5: NAÏVE BAYES CLASSIFIER

Naïve Bayes Classifier

- **Naïve Bayes classifier** is a **supervised** machine learning algorithm.
- It does not learn which features are important to differentiate between classes compared to discriminative classifiers.
- It belongs to **generative learning algorithms** family.
 - It seeks to model the distribution of inputs of a given class or category.
- In some domains its performance has been shown to be comparable to that of neural network and decision tree learning.
- It can be used for text classification.

Naïve Bayes Classifier

- Naïve Bayes **assumes** that **features are conditionally independent** (i.e. unrelated to any of the other feature in the model) given the target class.
- It also **assumes** that **all features contribute equally** to the outcome.
- This is not the case in real world.
- However, this assumption simplifies classification.
- Despite the unrealistic independence assumption, the classification algorithm performs well.
- **Example:** A fruit may be an orange if it is round, yellow, and about 10+ cm in diameter.
 - A naive Bayes classifier considers each of these features to contribute independently to the probability that this fruit is an orange
 - This classifier does not care whether there exist any possible correlations between the color, roundness, and diameter features.

Naïve Bayes Classifier

- Naïve Bayes assigns probabilities $p(C_k|x_1, \dots, x_n)$ for each of the K possible outcomes or classes C_k given a problem instance to be classified, represented by a vector $x = (x_1, \dots, x_n)$.
- By using Bayes theorem and expanding it with chain rule we get the formula to assign probability of belonging to class with label y:

Means proportional

$$P(y|x_1, \dots, x_n) \propto P(y) \prod_{i=1}^n P(x_i|y)$$

with only binary outcome:

$$y = \operatorname{argmax}_y P(y) \prod_{i=1}^n P(x_i|y)$$

Naïve Bayes Classifier

- **Example: Movie reviews**
- We want to classify each review as negative/positive
- We have a small dataset.

Review Text	Class
"I loved the movie."	Positive
"I hated the movie"	Negative
"A great movie. Good movie"	Positive
"Poor acting."	Negative
"Great acting. A good movie"	Positive

$$P(+) = \frac{3}{5}$$

$$P(-) = \frac{2}{5}$$

Naïve Bayes Classifier

- **Example: Movie reviews**
- Let's convert the data into a better representation.
- Use a **binary representation** where each word is a **feature**, and we indicate its presence or absence in the review.
- There are **10 unique words** in the review dataset.
- I, loved, the, movie, hated, a, great, poor, acting, good

Review Text	Class
"I loved the movie."	Positive
"I hated the movie"	Negative
"A great movie. Good movie"	Positive
"Poor acting."	Negative
"Great acting. A good movie"	Positive

Naïve Bayes Classifier

Label	Review	I	loved	the	movie	hated	a	great	poor	acting	good
	"I loved the movie."	1	1	1	1	0	0	0	0	0	0
	"I hated the movie"	1	0	1	1	1	0	0	0	0	0
	"A great movie. Good movie"	0	0	0	1	0	1	1	0	0	1
	"Poor acting."	0	0	0	0	0	0	0	1	1	0
	"Great acting. A good movie"	0	0	0	1	0	1	1	0	1	1

Naïve Bayes Classifier

Example: Movie reviews

- Positive and negative review probabilities
- $P(\text{positive}) = \frac{3}{5} = 0.6$
- $P(\text{negative}) = \frac{2}{5} = 0.4$
- Probabilities of word_i can be calculated given it is a positive/negative review with:
- $P(\text{word}_i|\text{positive}) = \frac{n_{\text{word}_i}+1}{n+|\text{vocabulary}|}$ or $P(\text{word}_i|\text{negative}) = \frac{n_{\text{word}_i}+1}{n+|\text{vocabulary}|}$

where n_{word_i} = the number of times word_i occurs in the positive/negative cases, n = the total number of words in positive/negative cases and $|\text{vocabulary}|$ is the size of vocabulary (count of unique words).

$$P(word_i|positive) = \frac{n_{word_i} + 1}{n + |vocabulary|}$$

$$P(word_i|negative) = \frac{n_{word_i} + 1}{n + |vocabulary|}$$

Naïve Bayes Classifier

Review Text	Class
"I loved the movie."	Positive
"I hated the movie"	Negative
"A great movie. Good movie"	Positive
"Poor acting."	Negative
"Great acting. A good movie"	Positive

Positive reviews

- $P(I|positive) = \frac{1+1}{14+10} = 0.0833$
- $P(the|positive) = \frac{1+1}{14+10} = 0.0833$
- $P(a|positive) = \frac{2+1}{14+10} = 0.125$
- $P(acting|positive) = \frac{1+1}{14+10} = 0.0833$
- $P(hated|positive) = \frac{0+1}{14+10} = 0.0417$
- $P(loved|positive) = \frac{1+1}{14+10} = 0.0833$
- $P(movie|positive) = \frac{5+1}{14+10} = 0.2083$
- $P(great|positive) = \frac{2+1}{14+10} = 0.125$
- $P(good|positive) = \frac{2+1}{14+10} = 0.125$
- $P(poor|positive) = \frac{0+1}{14+10} = 0.0417$

• Negative reviews

- $P(I|negative) = \frac{1+1}{6+10} = 0.125$
- $P(movie|negative) = \frac{1+1}{6+10} = 0.125$
- $P(poor|negative) = \frac{1+1}{6+10} = 0.125$
- $P(loved|negative) = \frac{0+1}{6+10} = 0.0625$
- $P(great|negative) = \frac{0+1}{6+10} = 0.0625$
- $P(the|negative) = \frac{1+1}{6+10} = 0.125$
- $P(hated|negative) = \frac{1+1}{6+10} = 0.125$
- $P(acting|negative) = \frac{1+1}{6+10} = 0.125$
- $P(a|negative) = \frac{0+1}{6+10} = 0.0625$
- $P(good|negative) = \frac{0+1}{6+10} = 0.0625$

Naïve Bayes Classifier

$$y = \operatorname{argmax}_y P(y) \prod_{i=1}^n P(x_i|y)$$

Example: Movie reviews

- Now we trained our classifier, let's classify a new review.
- New Review: "I hated the poor acting"
- Calculate both possible outcomes:
- $P(\text{positive}|\text{I hated the poor acting}) = P(\text{positive}) \times P(I|\text{positive}) \times P(\text{hated}|\text{positive}) \times P(\text{the}|\text{positive}) \times P(\text{poor}|\text{positive}) \times P(\text{acting}|\text{positive})$
- $P(\text{positive}|\text{I hated the poor acting}) = 0.6 \times 0.0833 \times 0.0417 \times 0.0833 \times 0.0417 \times 0.0833 = 6.03 \times 10^{-7}$
- $P(\text{negative}|\text{I hated the poor acting}) = P(\text{negative}) \times P(I|\text{negative}) \times P(\text{hated}|\text{negative}) \times P(\text{the}|\text{negative}) \times P(\text{poor}|\text{negative}) \times P(\text{acting}|\text{negative})$
- $P(\text{negative}|\text{I hated the poor acting}) = 0.4 \times 0.125 \times 0.125 \times 0.125 \times 0.125 \times 0.125 = 1.22 \times 10^{-5}$
- The higher value determines the class, so the new review is a negative one.